KaxaNuk
Sharing knowledge

INVESTMENT BOOTCAMP

GETTING THE DATA AND FEATURE ENGINEERING.



03

From your provider to an analyzed signal: the Data Curator, the Data Refinery and the Data Analyzer, with Golden-Flow beside you. Step 3 of the eight.

IN THIS SESSION

01 GET

02 REFINE

03 ANALYZE

04 WRAP-UP

FROM THE DATA TO AN ANALYZED SIGNAL.



1 GET

Your data, from your provider.

- Data Curator: four providers
- Your whole universe, in one download
- Your c_* features: trend and traded value

You leave with: your data

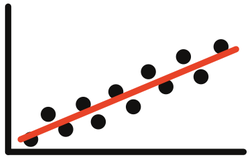


2 REFINE

Your universe, date by date.

- Data Refinery: reads it all
- Ranks and z-scores across stocks
- Your weight: the share of traded value

You leave with: your panel



3 ANALYZE

A signal, or just noise?

- Data Analyzer: explore your signal
- Against buy and hold
- The same strategy, another signal

You leave with: keep or drop



4 WRAP-UP

What you learned, and what comes next.

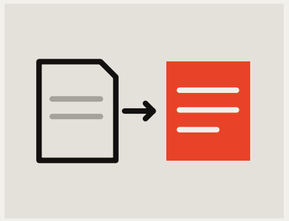
- The session's ideas, in your words
- Homework: finish your analysis
- Next: your portfolio, in S04

You leave with: your plan for S04

EIGHT STEPS. IN THIS SESSION, THE THIRD.

S02 | THIS SESSION | S04 AND S05 | S06 | BEYOND THE COURSE

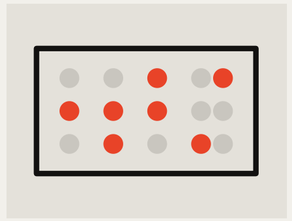
1 2 3 4 5 6 7 8



RESEARCH

S02

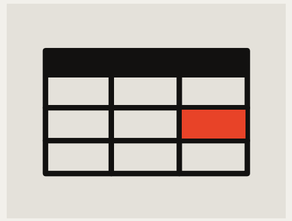
Your objective, from your library · OBJECTIVE.md



UNIVERSE

S02

Which companies and when; delisted included · Universe/

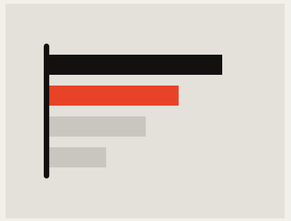


DATA

S03

Your features, and whether they carry a signal · Data/

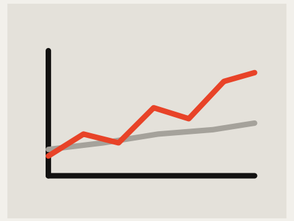
STEP 3



PORTFOLIO

S04

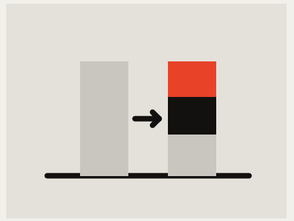
Your hypothesis, then the portfolios · Experiments/



BACKTEST

S05

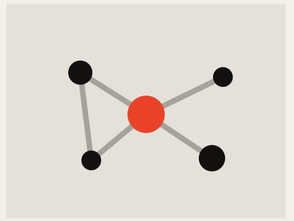
Your portfolio through time, net of costs



ATTRIBUTION

S05

Where the return came from



PAPER TRADING

S06

Tested without real money · Paper_Trading/



PRODUCTION

outside the course

From S02's homework

your chosen provider, its key in `Config/.env`, and your seed with the dead stocks.

THREE LIBRARIES, ONE STEP: YOUR DATA AND FEATURES.

CURATOR	REFINERY	ANALYZER
DATA CURATOR Downloads, validates and harmonizes data from four providers; your own calculations per instrument. **Get. Validate. Unify.**	**DATA REFINERY** Reads it all and builds features across stocks and through time: ranks, z-scores, sector neutral. **Enrich. Combine. Scale.**	**DATA ANALYZER** Explores your features and your signals: EDA, IC, charts and a simple backtest. **Explore. Detect. Understand.**

Feature engineering
from raw data to a signal worth testing, before any portfolio.

ONE CODE, ANY PROVIDER.

WHAT IT DOES	WHY IT MATTERS
Downloads from four providers: LSEG, FMP, Yahoo Finance and Sharadar.	You use the one you have.
The same tags for all: `m_`, `f_`, `fbs_`, `fis_`, `fcf_`, `d_`, `s_`.	Already mapped to each provider: you switch without touching your code.
Your own `c_*` calculations, over each instrument's history.	Simple features: moving averages, trend, traded value.
And `r_*` for the Data Refinery's calculations.	You know where each column comes from: `c_*` from the Data Curator, `r_*` from the Data Refinery.
Validates and harmonizes what arrives.	Catches a provider's outliers before they reach a signal.
One place for each API.	If a provider changes its API, it is updated once and everything works again: strategies, material, courses.
Open source.	It can grow: FRED, KaxaNuk's Analytics, Bloomberg, Intrinio, Databento, or your own databases.

LESS MAINTENANCE your code depends on the Data Curator, not on each provider's API.

USE THE ONE YOU HAVE. KNOW ITS GAPS.

Your provider

Each with its own identifier.

Yahoo Finance

No fundamentals, no dead stocks. To start; limited in a backtest.

no key

FMP

Fundamentals, dividends, splits.

a key · Basic, free

LSEG Workspace

VWAP; quarterly fundamentals only.

a subscription

Sharadar

Matches KaxaNuk's Analytics: the one in the course's slides.

paid

Your seed uses your provider's `main_identifier`: if you switch providers, check it.

ONE DOWNLOAD, YOUR WHOLE UNIVERSE.

ARYA · TREND-AND-VOLUME

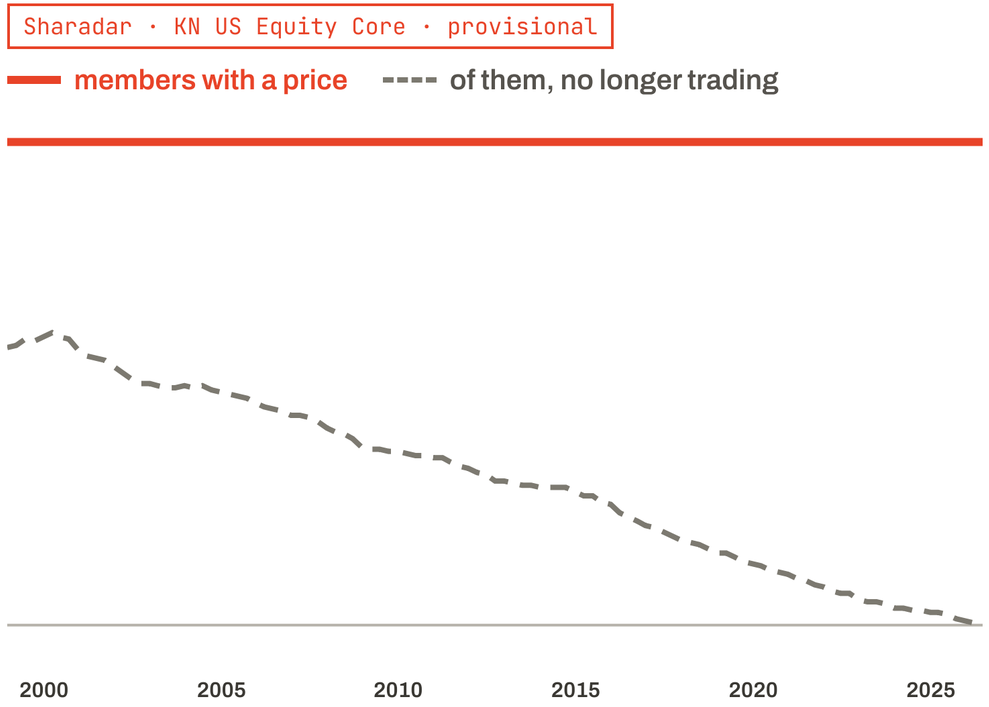
Download my whole universe, the dead stocks too. 1

With your seed and your provider, two passes: the second, only what was missed. Go ahead?

go ahead

Done: one file per stock. Now, the coverage. 2

COVERAGE · KN US EQUITY CORE · EVERY REVIEW



Every member of every review has a price. Of the 643 in 1998, more than half no longer trade: without them, your universe lies.

1. Your whole universe, the dead stocks too.
2. Then the coverage: from when it is usable.

YOUR FIRST FEATURES, PER INSTRUMENT.

DATA/CURATOR/

custom_calculations.py 1

c_sma_50	50-day average
c_sma_200	200-day average
c_trend_50_200	50 against 200
c_golden_cross	1 or 0: the cross
c_traded_value	VWAP × volume
c_traded_value_sma_63d	its 63-day average

ARYA · TREND-AND-VOLUME

Add the 50- and 200-day averages, their cross and the traded value. 2

I check the built-ins first. I propose six c_*, on the adjusted price. Go ahead?

go ahead

Written. Run the curator again: every stock. 3

1. c_*: one instrument's history.
2. Reuse first: the Features page.
3. A variant, like 63/252, is another c_*: another download.

`data-curator-custom-calculations`

FROM EACH STOCK TO YOUR WHOLE UNIVERSE.

WHAT IT DOES	AN EXAMPLE
Reads everything you downloaded	The Data Curator's files, and those of any API it does not have yet.
Compares across stocks, on each date	A rank or a z-score of your signal against the whole universe.
Sector neutral	Point-in-time classifications, to compare within each sector.
Weights among the eligible	Each stock's traded value over the total of its date's eligible stocks: Golden-Flow's weight.
Factor and signal metrics	Any calculation that needs the whole universe: N stocks by N.

BY DATE a rank is taken within its date; a value dated t reads up to t – 1.

YOUR WEIGHT, AMONG THEM ALL.

DATA/REFINERY/

`custom_calculations.py` 1

- `r_traded_value_share` its part, by date
  - `c_golden_cross` who is eligible
  - `c_traded_value_sma_63d` how much it trades

`Time_Series/` the panel

ARYA · TREND-AND-VOLUME

The share of traded value, among each date's eligible stocks. 2

Plan: by date, only with the cross at 1, summing to 1. Go ahead?

go ahead

It ran: `Data/Refinery/Time_Series/`. 3

1. Golden-Flow: its weight, in the refinery.
2. It needs every stock of the date: `r_*`.
3. The diff: by date? Does it sum to 1?

A SIGNAL, OR JUST NOISE?

WHAT IT DOES	WHAT FOR
Exploratory data analysis (EDA)	Distributions, outliers, coverage and stability: understand your data before using it.
Ready-made analyses and charts	IC, ICIR, ranks and deciles, on the platform or in code.
Statistics and machine learning	Several methods to measure whether a signal anticipates returns.
A simple backtest	The signal alone, with the Backtest Engine, before mixing it into the portfolio.
Your own notebooks	marimo or Jupyter: analyses to your measure, with comments.
Findings for the team	Documented and commented, to discuss them fast.

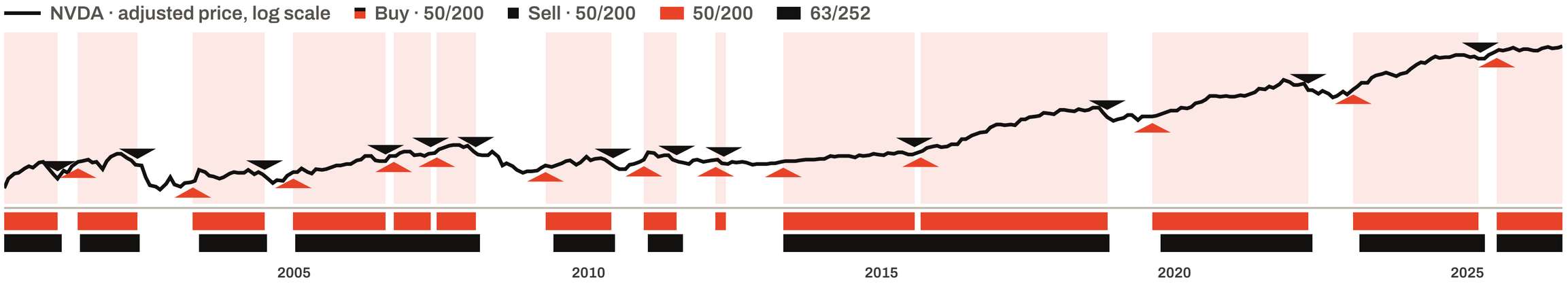
BEFORE THE PORTFOLIO a signal earns its place alone, or is dropped.

EXPLORE YOUR SIGNAL, STOCK BY STOCK.

SIGNAL EXPLORER · NVDA · 50/200 AND 63/252

Sharadar · to 2026-08-14

Stock: NVDA ▾ | Signals: 50/200, 63/252 ▾ | Log scale



Price adjusted for splits and dividends, at each month's close; each strip, the days its signal is at 1. From 2000-01-20 to 2026-08-14.

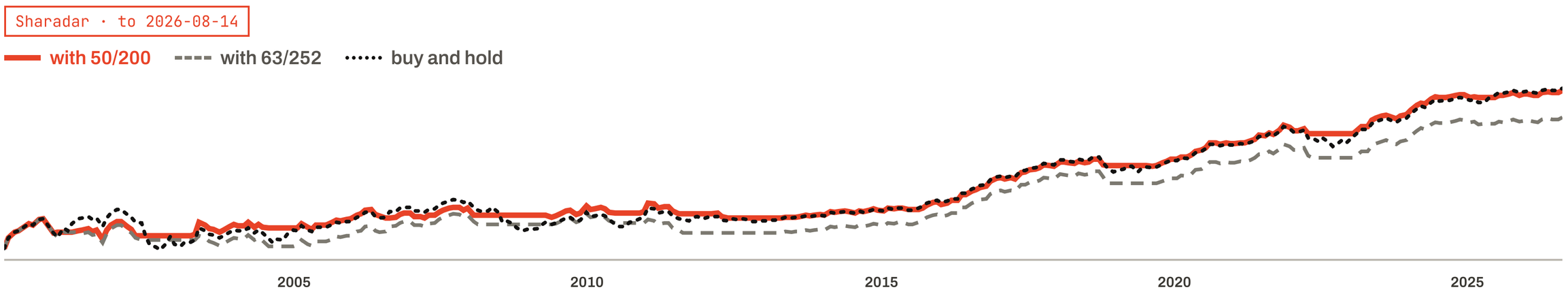
1 and 0
buy when it turns on, sell when it turns off; before 252 days, no signal.

One stock teaches
the rule; it does not decide. NVDA is a known winner.

HOW MUCH DID YOUR SIGNAL MAKE?

NVDA · GROWTH OF 1 · 50/200, 63/252 AND BUY AND HOLD

Sharadar · to 2026-08-14



MAX DRAWDOWN · 50/200	MAX DRAWDOWN · 63/252	MAX DRAWDOWN · HOLD
-70.0%	-76.8%	-89.8%

Log scale. Buy when it turns on, sell when it turns off, one day later, no costs. 1 became 2,323, 610 and 2,618.

The 50/200
came close to holding, with a far smaller drawdown.

The 63/252
came late to the rallies and fell behind.

50/200 OR 63/252? ONE STOCK AND TWENTY.

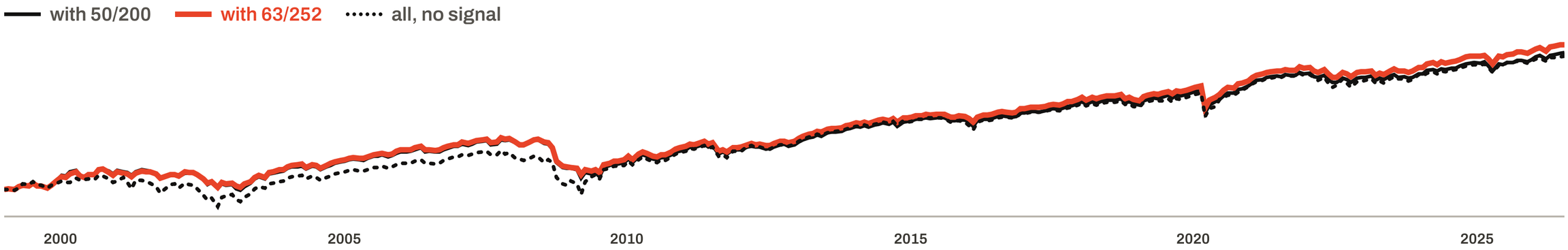
PER STOCK	50/200	63/252	BUY AND HOLD
NVDA · Sharpe	0.87	0.74	0.80
NVDA · max drawdown	−70.0%	−76.8%	−89.8%
Top 20 · best Sharpe	5 of 20	3 of 20	12 of 20
Top 20 · median Sharpe	0.57	0.53	0.56
Top 20 · median drawdown	−62.9%	−56.6%	−73.5%

AMONG THE LARGE holding usually wins on Sharpe; the signals cut the drawdown.

THE SAME STRATEGY, ANOTHER SIGNAL.

EQUAL WEIGHT ON THE CROSSED-ON · EVERY 21 DAYS · KN US EQUITY CORE

Sharadar · KN US Equity Core · provisional



SHARPE · 50/200	SHARPE · 63/252	SHARPE · ALL
0.62	0.67	0.54

Growth of 1, log scale, no costs; each date's members, the dead included.

Change one thing

same universe, weight and rebalancing; only the signal changes.

In a portfolio it helped

more Sharpe and less drawdown than all with no signal.

YOUR SIGNAL CANNOT KNOW THE FUTURE.

Look-ahead

Using what you did not know yet.

The price

Today's signal uses prices up to yesterday.

The report

An earnings figure counts from its release.

The adjustment

Only data that existed on that date.

The assistant

It knows what happened: figures from your files.

The same delay for every signal you compare.

WHAT WORKS BEST ACROSS ALL STOCKS?

1,919 KN US EQUITY CORE STOCKS · AVERAGE · SHARADAR

AVERAGE	50/200	63/252	HOLD	
Annual return	1.7%	2.0%	-0.6%	
Sharpe	0.17	0.19	0.25	
Sortino	0.29	0.32	0.39	
Calmar	0.08	0.09	0.07	
Max drawdown	-51.8%	-51.8%	-67.8%	
The 713 that fell	-7.4%	-6.9%	-23.8%	1
The 1,206 that rose	7.0%	7.4%	13.1%	2

1. Those that fell: the signal cut the loss to a third.
2. Those that rose: it cost half the return.

YOUR ANALYSIS, IN WRITING.

ARYA · TREND-AND-VOLUME

Compare 50/200 and 63/252 against holding, across my whole universe. 1

Explorer, a table per stock and the same strategy with each signal. Go ahead?

go ahead

Done, in RESULTS.md. Your decision is missing, with N = 2. 2

RESULTS.MD · EXAMPLE: THE KN US EQUITY CORE

SIGNAL	SHARPE	MAX DD	AGAINST ALL	DECISION
50/200	0.62	-50.9%	+0.08	
63/252	0.67	-49.1%	+0.12	
All, no signal	0.54	-55.4%	—	

The decision is yours: against the threshold you saved before looking; N = 2.

1. Ask for the full analysis.
2. Decide and write it, against your threshold.

`data-analyzer-runs`

THE SESSION'S IDEAS, IN YOUR WORDS.

01

Why does one code serve any provider?

02

When is a feature `c_*`, and when `r_*`?

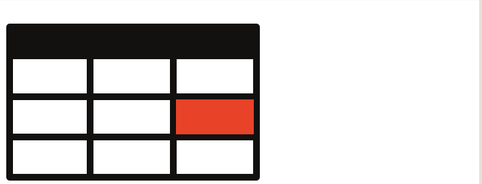
03

Why can a signal that makes less than holding still be useful?

04

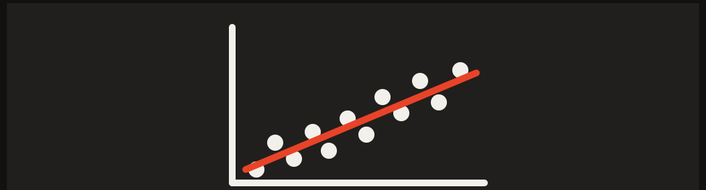
Why analyze a signal alone, before the portfolio?

BEFORE S04: ARRIVE READY.



01 YOUR DATA

- Your whole universe, the dead too.
- Check the coverage, and from when.
- Your `c_*`, with its report.
- Ask `next`.



02 YOUR SIGNAL

- Your `r_*`: your weight, by date.
- Explore it; compare it with holding.
- Its table, in the Analyzer.
- Keep or drop, in `RESULTS.md`.



03 READY FOR S04

- Your kept signal, with its table and N.
- Golden-Flow: its `RESULTS.md`.
- Golden-Flow: read its `BLUEPRINT_1.md`.
- How you would weight.

ANALYZED FIRST. BUILT NEXT.

Your data, your features and your signal, analyzed before any portfolio.

LEARN

The Bootcamp continues, then the pillars.

`github.com/KaxaNuk/Investment-Bootcamp`

TEST

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

TELL US

What your researcher helped you do.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

Session 04 · Write Your Hypothesis and Build the Historical Portfolios

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Nothing here is a recommendation to buy, sell or hold any security; backtests, Golden-Flow's included, are hypothetical, in sample and never traded, and their analysis is a measurement of past data, not a result; illustrative charts carry no data. Your researcher can be wrong; every decision is yours. What your researcher reads goes to the company behind your assistant, never to KaxaNuk.

Data providers' pages, FMP's plans, tools and works read 2026-10-11; tools checked again 2026-10-13; the figures, from Sharadar prices of the KN US Equity Core, whose methodology is still provisional, computed 2026-10-17, one day late and with no costs. Shared under CC BY 4.0; the KaxaNuk name, logo and isotipo are not.